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Thermodynamic Evidence of Structural Transformations in CO_2-Loaded Metal–Organic Framework $Zn(MeIm)_2$ from Heat Capacity Measurements

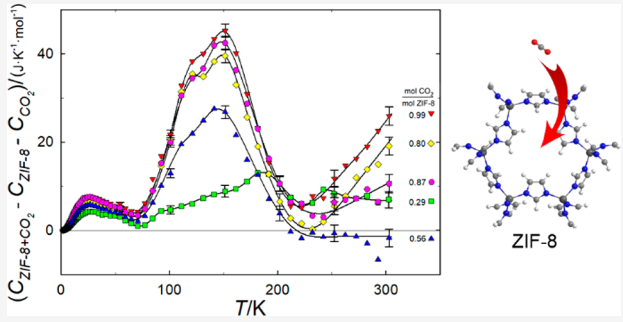
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ABSTRACT: Metal–organic frameworks are a class of porous compounds with potential applications in molecular sieving, gas sequestration, and catalysis. One family of MOFs, zeolitic imidizolate frameworks (ZIFs), is of particular interest for carbon dioxide sequestration. We have previously reported the heat capacity of the sodalite topology of the zinc 2-methylimidazolate framework (ZIF-8), and in this Article we present the first low-temperature heat capacity measurements of ZIF-8 with various amounts of sorbed CO_2. Molar heat capacities from 1.8 to 300 K are presented for samples containing up to 0.99 mol of CO_2 per mol of ZIF-8. Samples with at least 0.56 mol of CO_2 per mol of ZIF-8 display a large, broad anomaly from 70 to 220 K with a shoulder on the low-temperature side, suggesting sorption-induced structural transitions. We attribute the broad anomaly partially to a gate-opening transition, with the remainder resulting from CO_2 rearrangement and/or lattice expansion. The measurements also reveal a subtle anomaly from 0 to 70 K in all samples that does not exist in the sorbate-free material, which likely reflects new vibrational modes resulting from sorbate/ZIF-8 interactions. These results provide the first thermodynamic evidence of structural transitions induced by CO_2 sorption in the ZIF-8 framework.



INTRODUCTION

Metal–Organic Frameworks and Gate Opening. Metal–organic frameworks (MOFs) are a class of materials that have attracted increasing interest in recent years due to their wide variety of applications. MOFs are composed of metal ion nodes linked by organic molecules,[1−3] and their properties can be tuned by using a variety of different nodes and linkers.[3,4] MOFs have been proposed for use in numerous applications, such as catalysis,[5−11] drug delivery,[12−14] separation and molecular sieving,[15−22] sensing,[23−25] and gas sequestration.[26−31]

MOFs exhibit a high level of framework flexibility,[32] which enables many of their applications. One important example is gate opening, an effect that has been the focus of intense study. Gate opening is sometimes defined as a transition in structure from nonporous to porous,[32] but the term is often used more broadly to refer to any gate-like swinging of linkers that allows greater access to the pores in a framework.[33−35] Gate opening has been observed indirectly in a variety of MOFs, and it is primarily detected through the measurement of sorption isotherms.[36−47] Steps and hysteresis in sorption isotherms are attributed to structural changes induced by increased pressure or number of guest molecules, and simulations are often used to model these changes in the MOF structure. Note that while much of the literature uses the term "adsorption" to refer to the process of a MOF taking up a gas, we have chosen to refer to the process as "sorption" because in it the gas actually enters the MOF structure.

However, while sorption isotherms aid in identifying materials that exhibit gate opening, they provide limited information about its mechanism. Thus, further experimental investigation is needed to understand such transitions at the structural, molecular, and thermodynamic levels. Although such investigation is difficult due to the challenges of performing experiments under elevated gas pressures, researchers have employed a variety of techniques to understand the gate-opening phenomenon, including X-ray diffraction (XRD) either under variable pressure[43] or with guest molecules left over from synthesis;[44] enthalpy of sorption;[48] pressure-gradient differential scanning calorimetry;[43] inelastic neutron scattering (INS);[34,49,50] nuclear

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magnetic resonance (NMR);[51−53] and infrared microscopy (IRM).[54]

Gate Opening in ZIF-8. One of the most widely studied MOFs is the sodalite topology of the zinc 2-methlyimidazolate framework (zinc imidazolate framework 8, or ZIF-8), a commercially available MOF with a wide variety of applications.[8,55−58] ZIF-8 has been shown to allow molecules larger than its apparent pore apertures into the structure;[18] therefore, framework flexibility plays a key role in the sorption properties of ZIF-8. Gate opening of ZIF-8 was first reported by Moggach et al.[59] in 2009. They performed X-ray diffraction measurements on ZIF-8 in a diamond anvil cell using a methanol/ethanol solution as a hydrostatic medium to determine structural behavior under pressure. They observed a single-crystal to single-crystal phase transition at 1.47 GPa that was caused by ZIF-8 imidazolate linkers swinging and widening the channels into the pores. With the presence of a high-pressure phase confirmed, a key question became whether gas sorption could induce the same transition. Fairen-Jimenez and co-workers[36,38] studied the ability of gases to induce gate opening and found that the gate-opening transition observed by Moggach et al. is needed to adequately describe the sorption behavior of a variety of sorbates in ZIF-8, including those that do not show signs of gate opening (steps of increased sorption) in their sorption isotherms.

Additional experiments have further investigated the mechanism behind the gate-opening transition. Casco et al.[34] performed INS on ZIF-8 samples that were placed under pressurized N_2 gas at 77 K and then cooled to 10 K. Shifts in the measured INS spectra confirmed that the position of the linkers had changed. Casco et al. further measured spectra of ZIF-8 confined in nanocavities of activated carbon to determine whether the gate-opening mechanism involved an expansion of the lattice; the similarities between the spectra in and outside of activated carbon suggested that there was no significant lattice expansion in this case. However, this result conflicts with the findings of others[60−62] who suggested that lattice expansion is associated with gas sorption. In particular, Zhang et al.[62] performed XRD and Raman spectroscopy under isobaric N_2 atmospheric conditions over a range of temperatures and found two distinctive features that they attributed to lattice expansion (near 150 K) and a gate-opening transition (near 110 K). After their initial work, which claimed that lattice expansion does not occur, Casco and co-workers[49] later stated that gas sorption does in fact induce lattice expansion and that the expansion is accompanied by a reorientation of the linkers (i.e., gate opening) only when the sorbate has a strong quadrupole moment. The evidence of a sorption-induced lattice expansion is interesting because the transition is favored by pressure; a possible explanation of this seeming contradiction is that, upon sorption, the volume of the entire sorbate/MOF system decreases.

The detailed mechanism of the movement of methylimidazolate linkers during the gate-opening transition is unclear. Tanaka et al.[37] suggested that linkers have limited, gradual rotation under initial loadings until a point is reached where a gate-opening transition spontaneously occurs. In a similar vein, Chokbunpiam et al.[35] stated that the linkers gradually shift to restrict the pores at low loadings and rapidly rotate to open the pores at high loadings. In contrast, Coudert[63] proposed that the swinging of the imidazolate linkers was purely a gradual process. Note, however, that all three studies used different gases (Ar, CO_2, and N_2, respectively). Ania et al.[60] claimed that a gas-induced structural change of ZIF-8 occurs, but the change is not as significant as that of forming the high-pressure phase observed by Moggach et al.[59] In addition, crystal size has been shown to affect the sorption properties of ZIF-8, with smaller crystals requiring higher pressure to undergo a gate-opening transition.[46,64,65]

The transient movement of the linkers under lower pressures has also been a focus of study. Pantatosaki et al.[53] investigated diffusivities of CO_2 and CH_4 in ZIF-8 with pulsed-field gradient NMR and IRM and concluded that the swinging of the imidazolate linkers was important for allowing entry into the pores, although there was no permanent structural transition at the pressures investigated. This conclusion is supported by the work of Pusch et al.,[52] Ueda et al.,[66] and Kolokolov et al.,[51] who observed "flipping" of the methylimidazole linker between two sites that allows larger molecules to enter the ZIF-8 structure. Other studies[67,68] have shown that the framework flexibility of ZIF-8 has a significant impact on sorption kinetics, but claim that it does not play a major role in the sorption thermodynamics. Thus, even in the absence of a permanent gate-opening transition, transient gate opening is essential for describing the dynamic behavior of ZIF-8.

CO_2 Sorption in ZIF-8. The ZIF-8/CO_2 system has received attention across a wide range of pressures,[69,70] but it is more poorly understood than many other ZIF-8/sorbate systems as the result of two challenges that make it difficult to study structural phenomena. First, the diameter of CO_2 is smaller than the diameter of the channels into the ZIF-8 pores,[68] and therefore gate opening, transient or permanent, is not necessary to allow CO_2 molecules into the pores. In fact, simulations have shown that calculated CO_2 sorption is not particularly sensitive to the simulated flexibility of the framework,[35,36,68,71] although flexibility is necessary to correctly model diffusion.[68,72] Second, the relatively high sublimation point of CO_2 makes it more difficult to study at low temperatures, limiting the scope of some studies.[47,73] For these reasons, a CO_2-induced gate-opening transition in ZIF-8 has only been confirmed by simulation studies at very high pressures.[35] Because no direct experimental evidence of this transition has been presented, further investigation of this system is needed.

Low-temperature heat capacity measurements are a proven method for investigating structural phenomena. Because they give direct insight into the lattice dynamics of materials,[74] they are particularly well suited for studying a phenomenon such as gate opening, which depends on collaborative atomic motions in the framework. In addition, heat capacity measurements allow for direct observation of the material in bulk, as opposed to the limited scope of simulation studies or studies that are confined to measuring surface phenomena. To provide an additional experimental tool to study interactions between MOFs and sorbates, we have recently developed a method to measure low-temperature heat capacities under elevated gas pressures of up to 689 kPa.[75]

We previously performed low-temperature heat capacity measurements on empty ZIF-8[76] and observed no transitions in the heat capacity, which suggests that the structural flexibility of the framework alone is not sufficient to induce a gate-opening transition in the empty ZIF-8 structure. To investigate the potential for a sorbate to cause a gate-opening transition, we have measured the low-temperature heat capacity of ZIF-8 loaded with various amounts of CO_2. Our

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results provide the first experimental evidence of a CO_2-induced gate-opening transition at low temperature in ZIF-8. These results are analyzed and compared to those of previous studies to help clarify the mechanism behind framework flexibility and gate-opening effects in ZIF-8.

RESULTS AND DISCUSSION

Heat Capacity Measurements. The ZIF-8 samples in this study have been used in several previous studies, and details on the synthesis and characterization are given elsewhere;[76] TGA and FTIR-ATR measurements showed it to be free of solvents used in synthesis and 98% pure. Samples of ZIF-8 were prepared for heat capacity measurements by encasing them in copper foil (0.025 mm thick, 99.999% purity from Alfa Aesar) according to an established method,[75,77,78] with two copper coils embedded in the sample to ensure uniform heating.

To maintain CO_2 loading throughout the course of the measurement, the pellets of ZIF-8 were sealed under pressurized CO_2 inside small pressure cells that could be loaded into the calorimeter. Aluminum DSC pans from PerkinElmer served as the pressure cells, and they were sealed under various pressures of carbon dioxide using a commercial crimper inside of a pressure chamber. Apiezon N grease was applied to the top and bottom of the pellet prior to sealing to provide a thermal link with the aluminum pan. The method for sealing samples in a pressurized atmosphere and measuring their heat capacity is detailed elsewhere and has been previously characterized as giving results with accuracy comparable to that of standard measurements on samples not sealed in DSC pans.[75] The amount of CO_2 included with each sample was determined by buoyancy-corrected mass measurements performed before and after sealing. Each sample was loaded at either 689, 517, 345, or 207 kPa, but we found the loadings to be irreproducible based on pressure, perhaps due to variability in the crystal size of the samples. Details on the samples, including the masses of the aluminum, copper, and grease included in the measurement, are given in the Supporting Information.

The molar heat capacity of each sample was measured from 1.8 to 300 K via the thermal relaxation method[79] on a Physical Property Measurement System from Quantum Design. The heat capacity contributions of the sample holder, the calorimeter, and the grease thermally linking the pan to the sample holder were accounted for by performing an addenda measurement prior to the sample measurement, and the addenda heat capacity contributions were subtracted from the total.

The molar heat capacity of a sample of CO_2 sealed inside an aluminum DSC pan was also measured for comparison. To improve the sealing success rate, the base and lid of the DSC pan were deoxidized in room-temperature 20% NaOH for 1 min and then treated with 50% nitric acid for 5 min to remove byproducts of the deoxidation reaction from the surface of the aluminum prior to sealing. The pan was then sealed in 689 kPa CO_2, and the heat capacity of the pan and the sample was measured from 1.8 to 300 K according to the method detailed above.

To separate the heat capacity of the ZIF-8 and CO_2 from other contributions, we subtracted the heat capacities of the copper, the aluminum, and the grease inside of the pan using smooth functions representing each contribution. Details on these functions are given elsewhere.[75] A new aluminum function was required for the sample of CO_2 to account for the effect of the deoxidizing treatment on the pan. To prepare the function, we measured the heat capacity of a deoxidized DSC pan and fit the result using the method detailed in ref 73.

The measured molar heat capacities of the five samples of ZIF-8 (with various amounts of sorbed CO_2) are given in the Supporting Information, and a plot of the data is given in Figure 1. Data for the molar heat capacity of empty ZIF-8 from

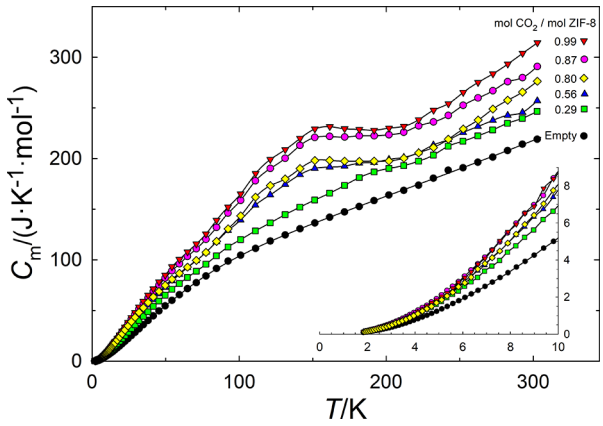


Figure 1. Molar heat capacity as a function of temperature for ZIF-8 loaded with 0, 0.29, 0.56, 0.80, 0.87, and 0.99 mol of CO_2 per mol of ZIF-8. Data for empty ZIF-8 are from ref 74. Empty depicted with ●, 0.29 mol of CO_2 depicted with green ■, 0.56 mol of CO_2 depicted with blue ▲, 0.80 mol of CO_2 depicted with yellow ◆, 0.87 mol of CO_2 depicted with pink ●, and 0.99 mol of CO_2 depicted with red ▼. A curve representing a theoretical fit is placed behind the data for empty ZIF-8; curves for other data sets are guides to the eye.

a previous measurement[76] are provided for reference. As expected, increasing amounts of CO_2 in the ZIF-8 samples correspond to a higher heat capacity. Samples with 0.56 mol of CO_2 per mol of ZIF-8 or more show a broad anomaly from 70 to 220 K. This anomaly does not appear in the data for unloaded ZIF-8, which suggests that it is caused by interaction between the ZIF-8 and the confined CO_2. The sample with 0.29 mol of CO_2 per mol of ZIF-8 appears to have a more narrow anomaly centered around 190 K, near the sublimation point of CO_2 under atmospheric pressure.[80] Comparison with a previous measurement[81] on the benzoic acid/nitrogen system, where no particular affinity between the gas and solid is expected and no anomalies except for the expected nitrogen phase changes appear, makes it clear that all observed anomalies are the result of special interactions between ZIF-8 and CO_2. Note that the broadness of each anomaly and the smoothness of the heating curves in the PPMS raw data files indicate that none of these anomalies are first-order transitions.

The measured molar heat capacity of CO_2 sealed in a pan without a sample pellet is given in the Supporting Information and plotted in Figure 2, with data for the molar heat capacity of solid CO_2 from Giauque and Egan[80] and Manzhelii et al.[82] provided for reference. The sublimation point of CO_2 appears as a large peak in the measured data near 210 K.

While heat capacity measurements are typically performed under constant pressure, the conditions required for studying the effects of gas sorption across a wide temperature range make a true isobaric measurement impossible.[75] Because the amount of CO_2 inside the pan, not the pressure, was held constant, we report our results as simply heat capacity (C), rather than constant-pressure heat capacity (C_p). When

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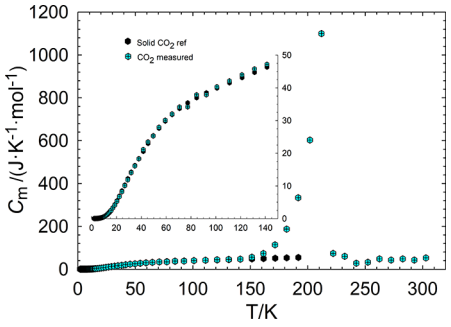


Figure 2. Measured molar heat capacity as a function of temperature for CO_2 sealed in a DSC pan, with reference data for solid CO_2 from refs 78 and 79. Measured data for CO_2 are depicted with cyan crossed ⬡, while reference data are depicted with ⬢. Inset shows data below 150 K.

expansion of the pressure cell is neglected, this heat capacity is approximately that at constant volume of the whole system (sample with sorbed CO_2 plus any free CO_2 in the cell). Our measurements also have added complexity due to competing effects on sorption: as temperature decreases, the capacity for CO_2 sorption increases,[83] the pressure of the CO_2 in the container drops, and the kinetics slow down. Therefore, we cannot predict the exact loading of ZIF-8 at any given temperature, nor can we calculate the pressure inside the pan. However, loading ZIF-8 with gas and then decreasing the temperature to 10 K was successfully used by Casco et al.[34] to show a structural transition, so we conclude that cooling a previously loaded sample is viable for our measurements as well. In addition, all but one of our samples lack a large anomaly near the normal sublimation point of CO_2, which indicates that at that temperature most of the free gas has already been sorbed into the sample. Because of the complexity of this system, we assumed that all of the CO_2 in each pan was sorbed into the ZIF-8, although there is likely some free gas near room temperature.

We also note that while the measurements reported here provide limited insight into possible kinetic effects related to the observed anomalies, analysis of the raw data suggests that our measurements reflect the equilibrium state of the system at each temperature. Each data point we report represents the average of three individual measurements, with 30−60 min between measurements in the range of the massive anomaly. Because the scatter of each trio of measurements is typical for what we observe for standard measurements on empty solids, it appears that a longer equilibration time would have no significant effect on the shape of the anomalies.

Isolation of the Heat Capacity Anomalies. To determine the source of the anomaly spanning the range from 70 to 220 K for all samples except the sample with the lowest loading, we prepared smooth functions representing the heat capacity of the empty ZIF-8 and of solid CO_2 and subtracted them from the total heat capacity to yield "excess" heat capacities resulting from interaction between sorbed CO_2 and the ZIF-8 framework. The CO_2 subtraction was prepared by fitting data for solid CO_2 from Giauque and Egan[80] and Manzhelii et al.[82] At temperatures above the sublimation point of CO_2, the data from Giauque and Egan were extrapolated using the theoretical function:

$$C = m \times D\left(\frac{\Theta_D}{T}\right) + n \times E\left(\frac{\Theta_E}{T}\right)$$

where $D(\Theta_D/T)$ and $E(\Theta_E/T)$ are Debye and Einstein functions, respectively, and m, n, Θ_D, and Θ_E are adjustable parameters. The ZIF-8 lattice subtraction was prepared by fitting data from a previous measurement[76] on ZIF-8 to theoretical functions. Theoretical fits were applied to low-temperature (<15 K) and high-temperature (>50 K) regions, and the two regions were connected using an orthogonal polynomial fit. Details on the fits for empty ZIF-8 are given elsewhere.[76]

A plot of the measured data minus the calculated heat capacity of solid CO_2 is given in Figure 3, and a plot of the measured data minus the calculated heat capacity of empty ZIF-8 (with the results given per mole of CO_2) is given in Figure 4.

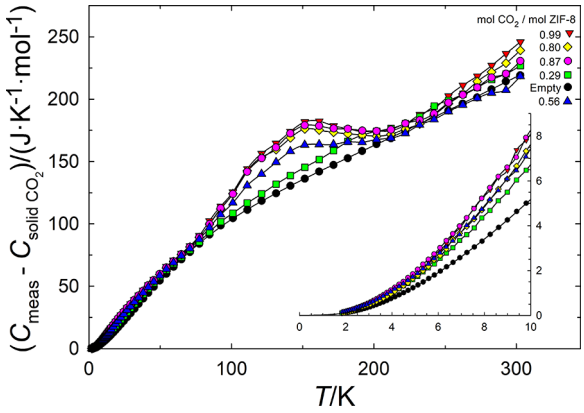


Figure 3. Measured molar heat capacity minus calculated heat capacity of solid CO_2 lattice for ZIF-8·XCO$_2$ (X = 0, 0.29, 0.56, 0.80, 0.87, and 0.99) as a function of temperature. Data for empty ZIF-8 are from ref 74. Empty depicted with ●, loadings of 0.29 mol of CO_2 depicted with green ■, 0.56 mol of CO_2 depicted with blue ▲, 0.80 mol of CO_2 depicted with yellow ◆, 0.87 mol of CO_2 depicted with pink ●, and 0.99 mol of CO_2 depicted with red ▼. Curves representing theoretical fits are placed behind the data for empty ZIF-8; curves for other data sets are guides to the eye.

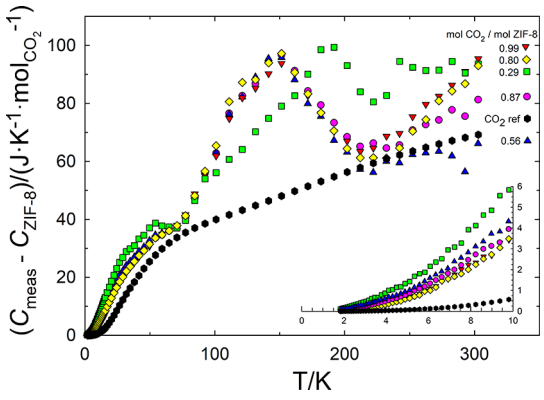


Figure 4. Measured molar heat capacity minus heat capacity of empty ZIF-8 for ZIF-8·XCO$_2$ (X = 0.29, 0.56, 0.80, 0.87, and 0.99) as a function of temperature. Results are given per mole of CO_2. Reference data for the heat capacity of solid CO_2 from refs 78 and 79, with values over 190 K extrapolated from data of ref 78, are given for comparison. Reference data are depicted with ⬢, 0.29 mol of CO_2 depicted with green ■, 0.56 mol of CO_2 depicted with blue ▲, 0.80 mol of CO_2 depicted with yellow ◆, 0.87 mol of CO_2 depicted with pink ●, and 0.99 mol of CO_2 depicted with red ▼.

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Neither subtraction eliminates the anomaly, which shows that it is characteristic of neither ZIF-8 nor CO_2. To further study the anomaly, we applied both the CO_2 and the ZIF-8 subtractions simultaneously to the data to isolate the excess heat capacity resulting from the ZIF-8/CO_2 interaction. We fit this result to a smoothed least-squares cubic spline and integrated the function to calculate the excess entropy and enthalpy associated with the possible transition for each sample. A plot of the data with both subtractions applied, along with spline fits of the data, is given in Figure 5. Error bars

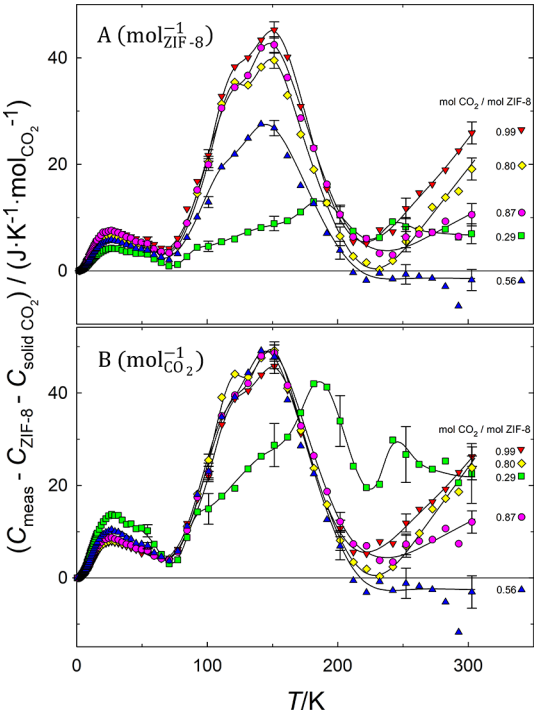


Figure 5. Excess heat capacity per mole of ZIF-8 (A) and per mole of CO_2 (B) as a function of temperature for ZIF-8·XCO_2 (X = 0.29, 0.56, 0.80, 0.87, and 0.99), obtained by subtracting both calculated ZIF-8 and CO_2 lattice heat capacities from the measured data. Data for 0.29 mol of CO_2 are depicted with green ■, 0.56 mol of CO_2 depicted with blue ▲, 0.80 mol of CO_2 depicted with yellow ◆, 0.87 mol of CO_2 depicted with pink ●, and 0.99 mol of CO_2 depicted with red ▼. Smoothed least-squares spline fits are placed behind the data.

indicating the measurement's resolution of 0.1% of the raw measured heat capacity demonstrate that the observed anomalies are resolvable. The calculated excess entropies and enthalpies at 298.15 K are given in Table 1.

Note that the large anomaly in the ZIF-8 data cannot be explained simply by sublimation of CO_2 in the pan. This is demonstrated by Figure 6, which compares the data in Figure 4 (heat capacity of ZIF-8·XCO_2 minus empty ZIF-8, given per mole of CO_2) and the measured heat capacity of CO_2 alone in an aluminum pan. The sublimation peak in the data for CO_2

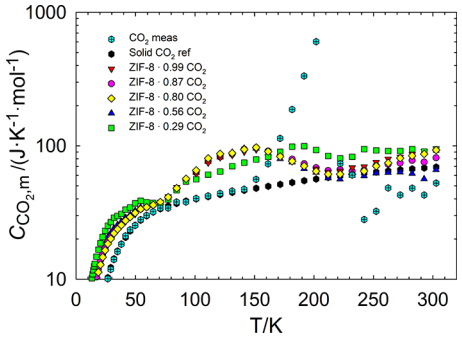


Figure 6. A comparison of the heat capacity values from Figure 2 (measured heat capacity of CO_2 in a pan) and Figure 4 (measured heat capacity of ZIF-8·XCO_2 minus that of empty ZIF-8, given per mole of CO_2). Measured data for CO_2 are depicted with cyan crossed ⬡, reference data for solid CO_2 depicted with ⬢, X = 0.29 mol of CO_2 depicted with green ■, 0.56 mol of CO_2 depicted with blue ▲, 0.80 mol of CO_2 depicted with yellow ◆, 0.87 mol of CO_2 depicted with pink ●, and 0.99 mol of CO_2 depicted with red ▼. Note the logarithmic scale on the y-axis.

alone takes place at a higher temperature and is far larger than any of the anomalies in the other data sets (note the logarithmic scale). Furthermore, were the peak due to sublimation, the temperature of the peak would require the pressure inside the pan to be quite low for CO_2 to sublimate (approximately 1.33 kPa), which would mean too little free CO_2 is available at that temperature to account for the size of the observed anomaly. On the other hand, the enthalpy associated with the anomalies (per mole of CO_2) is an order of magnitude smaller than the molar enthalpy of sublimation of CO_2, so the anomalies cannot reflect simply a modification of the temperature at which CO_2 sublimes. Thus, none of the observed transitions can be attributed to the sublimation of free gas.

Figure 5 reveals a subtle second anomaly between 0 and 70 K for all samples and highlights an apparent shoulder on the low-temperature side of the 70−220 K peak for the samples with higher loading, which suggests that this large peak may consist of two overlapping peaks. The excess heat capacity of the sample with 0.29 mol of CO_2 per mol of ZIF-8 is perhaps the most complex, with a kink near 90 K, a gradual rise until approximately 160 K, a significant anomaly from 160 to 220 K, and a smaller anomaly above 220 K. The behavior of the five data sets above 220 K is variable, but there is a correlation between the steepness of a high-temperature upturn and the loading pressure of the sample, except in the case of the 0.29 mol of CO_2 sample. Of the other four samples, those with the steepest upturn (0.99 and 0.80 mol of CO_2 per mol of ZIF-8) were loaded at the highest pressure (689 kPa), the sample with no upturn (ZIF-8·0.56CO_2) was loaded at the lowest pressure (345 kPa), and the sample with a moderate upturn (ZIF-8·0.87CO_2) was loaded at an intermediate pressure (517 kPa).

Table 1. Excess Entropy and Enthalpy of Anomalies for ZIF-8·XCO_2 at 298.15 K[a]

	ZIF-8·0.29CO_2	ZIF-8·0.56CO_2	ZIF-8·0.80CO_2	ZIF-8·0.87CO_2	ZIF-8·0.99CO_2
$\Delta^{298.15\ ^oC}S_m$/J K^{-1} mol^{-1}	57.62	46.29	43.25	44.67	41.18
$\Delta^{298.15\ ^oC}H_m$/J mol^{-1}	6605	4374	4420	4569	4245

[a]Calculated using spline fits of the data in Figure 5 from 0 to 300 K for X = 0.29 and from 0 to 230 K for other samples. Data given per mole of CO_2. Values have an estimated relative uncertainty of about 1%.

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The origin of the low T anomaly is likely a change in the phonon spectrum of the ZIF-8 framework due to CO_2/ZIF-8 interactions. The low temperature and broadness of the anomaly, along with its presence in the data for all samples of ZIF-8 loaded with CO_2, support this conclusion. Inelastic neutron scattering (INS) studies are currently underway to explore the vibrational density of states of this system.

Massive Anomaly from 70 to 220 K. The two overlapping peaks from 70 to 220 K confirm that there are structural transitions in the ZIF-8/CO_2 system despite the lack of steps in the CO_2 sorption isotherm, and normalizing the data for the amount of CO_2 present demonstrates that the size of these anomalies is essentially a function of CO_2 loading (see Figure 5B). The high-temperature onset agrees well with the data of Radhakrishnan and Narayana,[73] who observed an increase in the Brillouin shift of the ZIF-8/CO_2 system below 225 K. The presence of a transition also agrees with the observations of Russell and Migone,[47] who found that both isosteric heats and equilibration times for CO_2 sorption displayed maxima below the highest loadings, features that were correlated to transitions in isotherms of other gases. Because the sample with the lowest loading displayed a dramatically different heat capacity anomaly over the temperature range of the large transition, we conclude that there is a minimum loading required to induce the observed 70−220 K transition. The 0.29 mol of CO_2 sample will be discussed in detail separately below.

Furthermore, the presence of two overlapping peaks in the heat capacity suggests that the structural change of the ZIF-8/CO_2 is not one simple change, but rather two somewhat overlapping changes. The literature suggests three possible explanations of the changes: (1) rearrangement of the sorbate, (2) expansion of the ZIF-8 structure, or (3) swinging of the imidazolate linkers.

The first potential explanation for one of the overlapping anomalies is rearrangement of the CO_2. This is an attractive explanation because if one of the peaks can be attributed to a structural change like a gate-opening transition, it would be expected to also cause rearrangement of the sorbates to accommodate the new structure, as suggested by Ania et al.[60] However, the broadness of the peak is difficult to explain. If a rearrangement of CO_2 accompanied the rearrangement of the structure, the transition in the CO_2 would be expected to take place over a narrow temperature range. It may be that the various local environments inside the structure of the ZIF-8 cause CO_2 molecules to "freeze" and reorder at different temperatures, resulting in a broad peak.

The second possible explanation, lattice expansion, seems promising because several studies conclude that such an expansion can be caused by gas sorption in ZIF-8.[60−62] However, this expansion is typically ascribed to additional sorbates entering the ZIF-8 structure, and in our measurement the amount of gaseous CO_2 inside the pan but outside the ZIF-8 pellet at the temperature of the transition is expected to be insignificant. Therefore, an expansion of the lattice does not seem likely unless the driving force of lattice expansion is not the entry of new sorbate molecules, but rather interactions with the guest molecules already present.

The final possible explanation of the anomaly is the rotation of the imidazolate linkers to open the ZIF-8 structure. This is likely the cause of the low-temperature shoulder of our transition, and this conclusion is well supported in the literature. Casco et al.[49] suggested that a key component driving the gate-opening transition was the presence of a quadrupole moment in a sorbate. Because CO_2 has a quadrupole moment greater than that of N_2,[84] CO_2 should be more capable of inducing gate opening in ZIF-8 than should N_2. Zhang et al.[62] observed two changes in the structure of ZIF-8 with sorbed N_2: one near 110 K that they correlated with gate opening and another near 150 K that they correlated with lattice expansion. These temperatures line up well with the peaks of the two anomalies. In the same experiment, they observed an "intermediate phase", which they determined was simply a mix of the high-pressure and low-pressure phases of ZIF-8. The presence of an "intermediate phase" could help explain the broadness of the observed peaks in the heat capacity. The broadness of the peaks could also be explained by Coudert's[63] suggestion that the swinging of the ZIF-8 linkers is a gradual process rather than a sudden shift. Both of these explanations of peak broadness also explain the near-linear dependence of the entropy associated with the anomaly on CO_2 loading shown in Table 1; either the gate-opening transition occurs only in part of the structure at a time, with additional CO_2 increasing the proportion of the ZIF-8 that undergoes the transition, or the degree to which the linkers rotate is controlled by the amount of CO_2 present. Because very little CO_2 is entering the ZIF-8 structure at the temperature of the transition and CO_2 already in the structure should be relatively evenly distributed, Coudert's explanation seems most consistent with both the broadness and the CO_2 dependence of our observed anomaly. Because there is little free gas at the temperatures of the anomalies we observed, we conclude that the strain introduced by the sorbate molecules becomes more pronounced at low temperatures, and the molecules present are already sufficient to produce the gate-opening effect. The observation of a temperature-induced structural transformation is supported by the recent work of Gandara-Loe et al.,[85] who found that another ZIF, ZIF-4, did not display a breathing phenomenon at 298 K after CO_2 sorption, but did display a structural change when cooled to 5 K with CO_2 still sorbed into the structure. Overall, a gradual rotation of the imidizolate linkers at low temperatures is most consistent with both the literature and our results.

Thus, we conclude that the low-temperature shoulder of the large anomaly correlates with a gate-opening transition in the form of a gradual rotation of the imidizolate linkers that is only favorable at low temperatures. The peak of the anomaly may be attributed to either lattice expansion, rearrangement of the sorbate, or a combination of the two.

Behavior of the Heat Capacity of ZIF-8·0.29CO_2. The explanations above can also be applied to the 0.29 mol of CO_2 sample, although it is more difficult to completely account for its behavior. One possible explanation for the peaks centered near 190 and 240 K is that low loading changes the temperature at which the major anomaly present in the other data sets occurs, perhaps moving the two overlapping peaks to higher temperatures and spreading them out until they are more easily resolved. This explanation is suggested by the fact that normalizing the data for the amount of CO_2 present makes the size of the peak centered near 190 K nearly match that of the major anomaly in the other data sets (see Figure 5B). Thus, one or both of the larger anomalies in the 0.29 mol of CO_2 data may be attributed to rearrangement of the sorbate, expansion of the ZIF-8 structure, or swinging of the imidazolate linkers, as discussed above. Another possibility is that a gate-opening transition corresponds to the kink near

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90 K; assuming that lower temperature increases the strain on the ZIF-8 lattice created by the sorbate, it seems reasonable that a lower temperature would be required to create the necessary strain when fewer CO_2 molecules are present. Rearrangement of the CO_2 molecules or an expansion of the ZIF-8 lattice could also contribute to the gradual rise from approximately 90 to 160 K, as explained above. Whether the differences between the 0.29 mol of CO_2 data and the other samples reflect modifications of the phenomena present in the other samples or different phenomena entirely, it is clear that the nature of ZIF-8/CO_2 interactions changes significantly with loading and stabilizes somewhat at loadings higher than a threshold value between 0.29 and 0.56 mol of CO_2 per mol of ZIF-8.

CONCLUSIONS

We have measured the low-temperature heat capacity of ZIF-8 with varying amounts of sorbed CO_2. Measurements were performed from 1.8 to 300 K on five samples containing up to 0.99 mol of CO_2 per mol of ZIF-8. Two anomalies are apparent in the data, and neither is characteristic of either CO_2 or empty ZIF-8. All samples exhibit a subtle anomaly from 0 to 70 K, which is likely due to a change in the phonon spectrum. Samples with at least 0.56 mol of CO_2 per mol of ZIF-8 exhibited a massive anomaly between 70 and 220 K with a shoulder on the low-temperature side, which suggests two overlapping sorption-induced transitions in the ZIF-8/CO_2 lattice. We attribute one of the overlapping anomalies to a gate-opening transition that occurs at loadings higher than a threshold value between 0.29 and 0.56 mol of CO_2 per mol of ZIF-8, and the other to either expansion of the lattice, rearrangement of CO_2 molecules, or both. This constitutes the first thermodynamic evidence of a gate-opening transition in ZIF-8 induced by sorbed CO_2.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/jacs.9b13883.

PPMS calorimetric measurements and measured molar heat capacity values (PDF)

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Notes

The authors declare no competing financial interest.

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